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Imaging Anatomy of the Cranial Nerves

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INTRODUCTION

During the last two decades, advances in neuroimaging using high-resolution computerized tomography (CT) and magnetic resonance imaging (MRI) have enabled brain structures to be visualized in highly detailed ways that now enable normal findings to be detected routinely and nervous system pathology to be detected. Although in their early years, some medical students and residents avoid the difficulties of the cranial nerves (CNs) because of their complex anatomy, after graduation a broad community of health sciences professionals maintain their interest in the anatomy of the cranial nerves: specialists in neurology, neurosurgery, radiology, otolaryngology, ophthalmology, maxillofacial surgery, radiation oncology, and emergency medicine, and in related fields; nurses and physician assistants, students and practitioners of dentistry, speech pathology, audiology, physical and health education, and rehabilitation sciences.

This chapter integrates the perspectives of radiologists and neurosurgeons to assemble a practical imaging review combining the use of 128-slice dual-source multiplanar images from CT cisternography with 3 T MRI (Roldan-Valadez, Martinez-Anda, & Corona-Cedillo, 2014). For each CN we present information regarding origin, course, function, and a brief listing of the most common pathologies and syndromes affecting that nerve. The scope of this chapter is the clinical anatomy necessary for understanding the complex anatomy of the CNs; readers will find specialized texts presenting more detailed information about particular topics.

TECHNICAL CONSIDERATIONS FOR IMAGING THE CRANIAL NERVES

Radiologists and clinicians can take advantage of the paired disposal of the CNs; therefore, it is possible to compare the size, signal intensity, and enhancement patterns of both nerves. The coronal plane might be best suited to study the cranial nerves I to VI, as they have a dominant

posteroanterior course; however, we recommend axial oblique planes. Cranial nerves VII to XII run in an antero-lateral-caudal direction but the lateral component is dominant. Hence, the best plane could be multiplanar reconstructions using axial oblique planes. Current 3D sequence algorithms allow multiplanar reconstructions for all cranial nerves, so they can be imaged in at least two planes (Borges & Casselman, 2007).

CT cisternography is currently used to image the CNs in a 128- or 256-slice dual-source CT scanner; it is recommended to use multiplanar reconstructed images of 1.2 mm or less thickness. MR images are nowadays acquired in 3 T scanners; this modality encompasses several challenges as the sequence choice, available high-resolution head coils, slice thickness, in-plane resolution, the use of special techniques/sequences like parallel imaging, asymmetric k-space, fat suppression, and so forth, will all influence the final image quality (Borges & Casselman, 2007). In this chapter, we present a summary of the most common MR sequences used in CN imaging (Table 14.1).

GENERAL REMARKS ABOUT THE PATHOLOGY OF THE CRANIAL NERVES

A wide range of disease processes can affect cranial nerves: primary, intrinsic to the nerve, or secondary. It is not uncommon for CN dysfunction to be the first sign of a neighboring disease owing to compression, perineural spread, or metastasis. Although it is not within the scope of this chapter to present a detailed description of the pathological conditions related to each CN, we list the most common clinical conditions affecting CNs (Table 14.2), the most common clinically affected anatomical sites related to them (Table 14.3), and the most relevant intracranial artery-CN relationships (Table 14.4) with clinical significance as they may require a proper assessment using MRI or CT imaging. We believe this additional information could be helpful in clinical practice.

TABLE 14.1 Imaging Characteristics of MR Sequences for Cranial Nerve Imaging

Clinical Interest	MR Sequence
Cisternal course of cranial nerves, assessment of neurovascular compression	3D FIESTA
Perineural spread tumor, meningeal infiltration, metastasis detection, and patterns of enhancement	Post gadolinium multiplanar T ₁ -w with/without fat saturation
Morphology of CFS spaces, nerve degeneration, lesion characterization	Multiplanar T ₂ -w with/without fat saturation
Vascular lesions, vascular loops, pulsatile tinnitus	MR angiography/venography
Parenchymal hemorrhage, hemosiderin detection	Axial MPGR
Anatomical definition, infiltration of fat planes, loss of normal marrow signal, parotid gland lesions	Multiplanar T ₁ -w
Globe pathology	3D FIESTA multiplanar T ₂ -w with/without fat saturation

Abbreviations: 3D, three-dimensional; CSF, cerebrospinal fluid; FIESTA, fast imaging employing steady-state acquisition; MPGR, multiplanar gradient-recalled; T₂-w, T₂-weighted imaging; T₁-w, T₁-weighted imaging.
Modified from Binder, Sonne, and Fisschbeln (2010)

TABLE 14.2 Most Common Clinical Conditions Affecting Cranial Nerves

Cranial Nerve	Pathology	Anatomical Site Affected
I	Nasal cavity neoplasms	Cribriform plate
	Olfactory groove meningiomas	Cribriform plate, olfactory bulb
II	Optic nerve inflammatory process	Optic nerve
	Optic nerve glioma	Intraorbital optic nerve
	Sellar region neoplasms	Optic chiasma
	Clinoid meningiomas	Intracranial optic nerve
	Cranial hypertension	Intracranial optic nerve
III, IV, VI	Temporal uncal herniation	Cisternal segment of CN III
	Cavernous sinus pathology	Intracavernous nerves segments
	Internal carotid aneurysms	Cisternal segment of CN III
	Clinoid meningiomas	Cranial nerves passing through superior orbital fissure
	Cranial hypertension	Cisternal segment of CN VI
V	Schwannomas	Trigeminal ganglia
	Trigeminal neuralgia	Arterial contact at root entry zone
VII, VIII	Pontocerebellar angle neoplasms	Cisternal segment at pontocerebellar angle cistern
	Middle and inner ear inflammatory and infectious process	Intra-petrous portion of both nerves
	Hemifacial spasm	Arterial contact at cisternal segment of CN VII
IX, X, XI, XII	Lower clivus and cranio-cervical junction neoplasms	Cisternal segment of nerves Cranial nerves passing through foramen yugulare or hypoglossal canal
	Osseous cranio-cervical junction pathology	Cisternal segment of nerves
	Cervical neoplasms or abscess	Exocranial cranial base

TABLE 14.3 Most Commonly Clinically Affected Anatomical Sites Related to Cranial Nerves

Anatomical Site	Cranial Nerves Affected
1. Cribriform plate	Olfactory nerve
2. Sella turcica	Optic nerve
3. Cavernous sinus	Oculomotor nerve Trochlear nerve Trigeminal nerve Abducens nerve V1 and V2 of trigeminal sinus
4. Superior orbital fissure	Oculomotor nerve Trochlear nerve V1 of trigeminal nerve Abducens nerve
5. Internal carotid artery	Optic nerve Oculomotor nerve
6. Anterior clinoid process	Optic nerve
7. Perimesencephalic cisterns and tentorial incisura	Oculomotor nerve Trochlear nerve
8. Cerebellopontine angle	Facial nerve Vestibulocochlear nerve
9. Petrous bone	Trigeminal nerve Facial nerve
10. Clivus	Trigeminal nerve Abducens nerve Glossopharyngeal nerve Vagus nerve Accessory nerve
11. Middle and inner ear	Facial nerve Vestibulocochlear nerve
12. Jugular foramen	Glossopharyngeal nerve Vagus nerve Accessory nerve
13. Craniocervical junction	Glossopharyngeal nerve Vagus nerve Accessory nerve Hypoglossal nerve

Olfactory Nerve

Function

The primary function of the olfactory nerve is to transmit olfactory impulses from the olfactory epithelium of the nose to the brain; it is better referred to as the olfactory bulb and tract and is an outgrowth of the forebrain. The primary sensory neurons are bipolar and are confined to the olfactory epithelium (Binder et al., 2010).

Anatomy

The molecules released into the environment reach the specialized sensory cells in the neuroepithelium in the back of the nasal cavity, where they are embedded in an area that

covers a region of about 5 cm² in size, and they are interspersed with glia-like supporting cells above a basal layer of stem cells. These cells are short-lived and continuously replaced by the basal stem cells. The olfactory neurons project cilia with specific receptors for odorants coupled with transduction machinery that amplifies the signal and generates action potentials in the neuron (Roldan-Valadez et al., 2014). Numerous olfactory nerves pass through the cribriform plate and dura mater from the olfactory epithelium in the sphenoethmoidal recess (Belluscio, Lodovichi, Feinstein, Mombaerts, & Katz, 2002); these nerves end in the olfactory bulb, which is situated over the cribriform plate; here the sensory axons synapse with mitral, tufted, and periglomerular neurons in units called glomeruli, which project through the

TABLE 14.4 Intracranial Artery-Cranial Nerve Relationships with Clinical Significance

Vascular Structure	Cranial Nerves Related
Anterior communicating artery	Olfactory nerve Optic nerve
Carotid artery	Optic nerve Oculomotor nerve Trochlear nerve Abducens nerve
Posterior communicating artery	Oculomotor nerve
Basilar artery	Oculomotor nerve Trochlear nerve Trigeminal nerve
Vertebral arteries	Glossopharyngeal nerve Vagus nerve Accessory nerve Hypoglossal nerve
Superior cerebellar artery	Trigeminal nerve
Anterior inferior cerebellar artery	Facial nerve Vestibulocochlear nerve
Posterior inferior cerebellar artery	Glossopharyngeal nerve Vagus nerve Accessory nerve Hypoglossal nerve

olfactory tract to the brain cortex (Ressler, Sullivan, & Buck, 1993). The olfactory bulb is the origin of the olfactory tract that travels in the olfactory cistern, situated ventral to the olfactory sulcus on the basal surface of the frontal lobe, above the orbital plate of the frontal bone. The olfactory tract divides in the lateral and medial olfactory striate forming the anterior boundaries of the anterior perforate substance (Turner, Gupta, & Mishkin, 1978). The lateral olfactory striate ends in the pyriform, the periamygdalar area and entorhinal cortex (primary olfactory cortex) (Stevens, 1969). The medial striate ends in the septal area (Monkhouse, 2006). It is interesting to note that this information does not reach the cortex through the thalamus.

Anatomical Landmarks

- Cribriform plate
- Basal surface of frontal lobes
- Anterior perforate substance
- Uncus of temporal lobes

Figure 14.1 shows some key MR and CT cisternography landmarks for CN I.

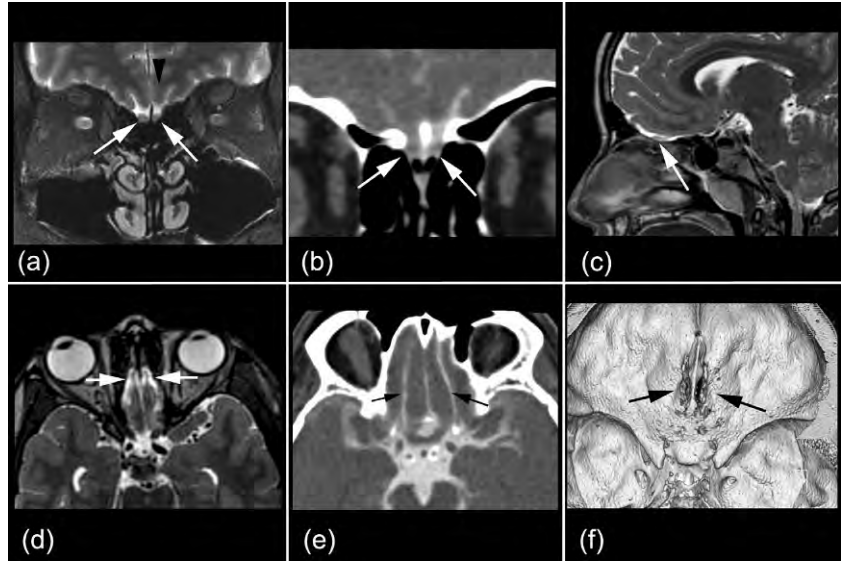
Clinical Correlation

The olfactory epithelium is affected in tumors arising in the top of the nasal cavity, like epithelial cell tumors,

angiofibromas, and esthesioneuroblastomas; this last tumor is an uncommon malignant neoplasm of the nasal vault, believed to arise from the olfactory epithelium (Dulguerov, Allal, & Calcaterra, 2001). Lesions of the olfactory pathway result from trauma (e.g., skull fracture) and often from olfactory groove meningiomas. These lesions cause ipsilateral anosmia (localizing value). Lesions that involve the parahippocampal uncus can cause olfactory hallucinations (uncinate fits, with déjà vu). The Foster Kennedy syndrome consists of ipsilateral anosmia, ipsilateral optic atrophy, and contralateral papilledema. It is usually caused by an anterior fossa meningioma (Fix, 2001).

Surgical management of lesions to the roof of the nasal cavity and anterior cranial base are difficult and technically demanding. There are two main ways of reaching this region: by craniofacial approaches and by endoscopic approaches. In the craniofacial route it is required to make a bifrontal craniotomy and remove the nasal bones and medial aspects of orbital rims. In some cases, these approaches are extended to the maxillary bone by a midfacial approach. Endoscopic approaches are reached by the transnasal route, and are classified as transfrontal and transcribiform in the sagittal plane, and supraorbital and transorbital in the coronal plane. These approaches require a large surgical learning curve, and have high risks of morbidity (Kassam, Snyderman, Mintz, Gardner, & Carrau, 2005).

FIGURE 14.1 Olfactory nerve. Coronal (a) T₂-w MR and (b) CT cisternogram images demonstrate bilateral and symmetric olfactory bulbs (*white arrows*) surrounded by CSF in the subarachnoid space. (c) Sagittal T₂-w MR reveals a normal olfactory bulb (*white arrow*) lying above the cribriform plate. (d) Partial view of the olfactory bulbs (*white arrows*) in an axial T₂-w MR. (e) Note the normal morphology of the olfactory sulcus (*black arrows*) in an axial plane of the CT cisternogram. (f) 3-D reconstruction of the anterior cranial fossa depicting numerous olfactory nerves passing through the cribriform plate (*black arrows*).



Optic Nerve

Function

The optic pathway transmits visual impulses from the retina to the brain. The optic pathway between the eyeball and the optic chiasm is erroneously called the optic nerve, but it is also an overgrowth of thalamic structures. The primary sensory neurons are bipolar and are confined to the sensitive epithelium of the retina (Dacey, 1996); the optic pathway originates in secondary sensory neurons, also located in the retina.

Anatomy

The optic nerve is divided into (1) the intraocular part, which leaves the eyeball to form (2) the intraorbital part, which passes through the medial part of the annular tendon and below the levator and superior rectus muscles; (3) the intracanalicular part, which passes through the optic canal, in the upper part of the sphenoid sinus, in front of the sella turcica, medial to the anterior clinoid process; and (4) the intracranial portion of the nerve, which is directed posteriorly, superiorly, and medially toward the optic chiasm, medial to the final segment of the internal carotid artery, crossing in an anteroposterior direction (Rhoton, 2002c). The intracranial portion of the optic nerve is 10–15mm long; the intracanalicular portion, 10–12mm; and the intraorbital portion, 25–30mm (Maroon & Kennerdell, 1984).

The optic canal opens into the superomedial corner of the orbital apex at the junction of the roof and medial wall. It is situated at the junction of the lesser wing with the body of the sphenoid bone, and is separated from the orbital fissure by the optic strut. The intracanalicular and intraorbital portions

are surrounded by dura and arachnoid. The subarachnoid space surrounding the intracranial part of the nerve extends forward from, and communicates with, the subarachnoid space around the intracanalicular and intraorbital portions of the nerve. The dural sheath around the optic nerve blends into the periorbita at the anterior end of the optic canal (Rhoton, 2002c). This relationship with meninges is important and explains changes of its appearance by fundoscopy in relation to intracranial pathology, like papillary edema seen in intracranial hypertension.

The optic nerve ends at the optic chiasm. At this point, the optic pathway is situated at the junction of the anterior wall and floor of the third ventricle, in posterior relation to the pituitary stalk (Rhoton, 2002d). The relationship of the optic nerve and the chiasm with vascular and neural structures in this region is very important, and explains the visual symptomatology caused by lesions in this area. There has been discussion of the existence of fibers from the nasal optic nerve that forms a loop into the contralateral optic nerve within the optic chiasm in a structure called the Wilbrand's knee, and that its injury causes a distinct visual field defect characterized by an ipsilateral central scotoma with contralateral superotemporal visual field defect, but there is no solid evidence of its presence in humans and anatomical studies are needed (Lee, Tobias, Kwon, Sade, & Kosmorsky, 2006).

The optic pathway continues as the optic tract, and 90% ends at the lateral geniculate nucleus, which has six cellular layers (Hendry & Reid, 2000), taking a posterolateral direction around the cerebral peduncles in relation to the tentorial incisura and the mesial surface of the temporal lobes. The visual projections end in one of two pathways in the lateral geniculate nucleus, in a magnocellular

pathway or in a parvocellular pathway, which convey different information to the visual cortex; such specialization is necessary for perception of color vision and spatial and temporal resolution (Kandel, 2013).

The rest of the visual projections reach the lateral geniculate nucleus, but some attain the superior colliculus of the quadrigeminal plate in the midbrain. This area also receives cortical and multisensory projections and forms part of eye movement's circuitry. Also, the pretectum of the midbrain receives inputs from retinal cells and controls pupillary reflexes (Kandel, 2013).

Finally, it ends as the optic radiations, reaching the primary visual cortex at the calcarine sulcus in the occipital lobe (Carroll & Wong-Riley, 1984). The optic radiations are part of the retrolenticular and sublenticular portions of the internal capsule; Meyer's loop is the anterior part of the optic radiation, located 28–34 mm from the temporal pole, reaching the anterior limit of the temporal horn's roof (Fernandez-Miranda et al., 2008).

The visual system has the most complex neural circuitry of all sensory systems. It reaches the primary visual cortex, called visual area 1 (V1) or striate cortex, because of a prominent stripe of white matter in layer 4, the stria of Gennari; it is located in Brodmann's area 17, and receives information exclusively from the contralateral half of the visual field. This cortex is about 2 mm thick and consists of six layers of cells. The principal layer for inputs from the lateral geniculate nucleus is layer 4. Here the visual inputs continue being transformed into perception of information by neurons organized in columns about 30–100 μm wide and 2 mm deep linked by horizontal connections with other columns, and with other cortical areas for visual integrative functions, such as sense of form, motion, depth, and so forth. (Kandel, 2013).

Vascular supply of the retina is derived from the ophthalmic artery, a branch from the internal carotid artery after the artery enters the subarachnoid space. Branches of the ophthalmic artery supply the optic nerve. As it approaches the chiasm, perforating branches from internal carotid and anterior cerebral arteries supply the visual pathway. The optic tract receives its supply from posterior communicating and posterior cerebral arteries. Lateral geniculate body receives irrigation from anterior choroidal and lateral posterior choroidal arteries. Optic radiations receive irrigation from the middle cerebral artery on its superior portion, and the posterior cerebral artery on its inferior portion. The primary visual cortex receives its supply from the posterior cerebral artery by one of its terminal branches, the calcarine artery (Rhoton, 2002b).

Anatomical Landmarks

- Eyeball
- Optic canal
- Anterior clinoid process
- Internal carotid artery
- Floor of the third ventricle and pituitary stalk
- Pituitary gland, sella turcica, sphenoidal sinus
- Cerebral peduncles
- Temporal horn of lateral ventricle
- Calcarine sulcus

Figure 14.2 shows some imaging landmarks for CN II.

Clinical Correlation

Given the broad areas related to the optic pathway, its lesions are classified according to the anatomy described earlier and their clinical consequences. The approach to a

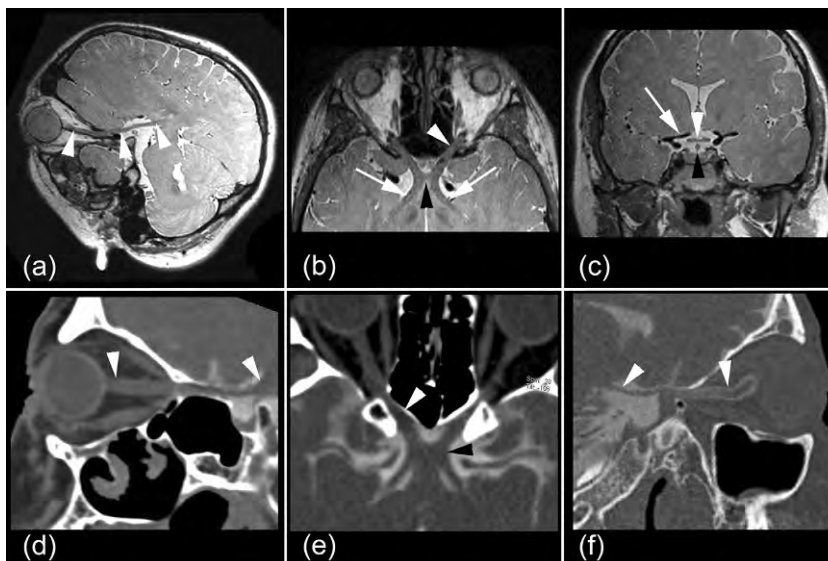


FIGURE 14.2 Optic nerves, optic chiasm, and optic tracts. (a) Sagittal curved T₂-w MR and (d, f) sagittal curved CT cisternography images illustrating the path of the optic nerves and the continuity among its segments. (b, e) Axial T₂-w MR and axial CT cisternography depict the optic nerves traversing the optic canal (white arrowheads) to meet at the optic chiasm (black arrowheads); the cisternal segment of the optic tracts is also evident (white arrows). (c) Coronal T₂-w MR demonstrates a normal optic chiasm (white arrowhead) beneath the anterior cerebral arteries (white arrow) and above the pituitary stalk (black arrowhead).

patient with visual symptoms starts with a detailed clinical evaluation. This includes techniques that allow identification of impaired visual perception or objective deficits. Impaired visual perception includes visual acuity, impaired color recognition, contrast discrimination, and visual field defects. Objective evaluation includes retinal changes seen with an ophthalmoscopy and an evaluation of pupillary reflexes (Paul, Brazis, & Biller, 2007).

Visual acuity, that is, the capacity for visual discrimination of the fine details of high contrast, reflects the function of the macular region, which is most often impaired by changes in the refractory characteristics of the eye. Once this impairment is excluded, it can be accepted that changes in visual acuity are secondary to lesions in the macular region in the retina or its projection in the visual pathway. Unilateral optic nerve lesions and chiasmatic lesions can impair visual acuity. Unilateral lesions of the retrochiasmatic visual pathway do not impair visual acuity. Impaired contrast sensitivity has localizing significance that is similar to that of impaired visual acuity. Color perception is often affected in areas of the visual field that correspond to partial field defects and in macular dysfunction. Color vision loss usually parallels visual acuity loss, and also occurs with lesions in the posterior visual pathways (Paul et al., 2007).

The shape and distribution of visual field loss is a valuable localizing sign. The most important feature for lesion localization is to note whether field defect is monocular, localizing the lesion in the retina or in the optic nerve, or binocular, localizing the lesion beyond the optic chiasm. Bitemporal field defects are most often due to a compressive mass lesion affecting the optic chiasm, such as lesions in the sellar area. Binasal hemianopias and quadrantanopias may occur, are usually asymmetric, and are usually due to bilateral intraocular disease of the retina or optic nerve, or may occur with hydrocephalus. Homonymous hemianopias appear with lesions in the retrochiasmatic pathways, in the optic tract, lateral geniculate body, optic radiations, or occipital lobe. Superior homonymous quadrantic defects may result from a lesion in the temporal (Meyer's) loop of the optic radiations. Involvement of the optic radiations in the depth of the parietal lobe gives rise to an inferior quadrantic defect. Medial occipital lesions cause homonymous hemianopias with macular sparing.

Once a detailed clinical and ophthalmoscopic evaluation has been made, investigation of the cause of the visual deficit continues with imaging and neurophysiological studies. MRI and CT studies are helpful in localizing lesions in the optic nerve, optic chiasm, and retrochiasmatic pathway. This can be complemented with biopsy, lumbar puncture, electroretinography, optical coherence tomography, visual evoked potentials, and certain antibody tests (Lueck, 2010).

Lesions that can affect the visual pathway are classified by the localization (anterior or posterior) and by the timing

of the visual loss. Anterior lesions are localized in the retina and optic nerve; acute lesions are most commonly vascular in origin, for example, retinal artery occlusion, acute ischemic optic neuropathy, and so on. Subacute lesions comprise vascular, autoimmune, and infective disease, such as typical idiopathic demyelinating optic neuritis, autoimmune retinal disease, and idiopathic intracranial hypertension. Chronic lesions include degenerative, compressive, toxic, nutritional, and infective/inflammatory diseases (Lueck, 2010).

In the case of posterior lesions, acute affection comprises vascular diseases (particularly stroke) and pituitary apoplexy. Subacute disease includes vascular lesions, hydrocephalus, demyelinating diseases, encephalitis, infections, and some tumors. Chronic diseases include principally tumors and some neurodegenerative diseases (Lueck, 2010).

Surgical approaches to the optic nerve include the trans-orbital and transcranial routes. The optic chiasm can be approached by the pterional cranial and endonasal trans-sphenoidal paths. The optic tract can be approached by the subtemporal and transchoroidal routes. The primary visual area at calcarine sulcus can be reached by the posterior interhemispheric approach.

Oculomotor Nerve

Function

The third cranial nerve has two main functions: (1) it innervates the extraocular muscles through somatic efferent pathways (except the superior oblique and lateral rectus muscles), being the final pathway of the complex mechanisms controlling ocular movements; and (2) it innervates the constrictor pupillae and ciliary muscles through parasympathetic components, causing constriction of the pupillae by luminous and accommodation stimuli, and the accommodation reflex of the ocular lens.

Anatomy

For each cranial nerve of the brain stem there is a real and an apparent origin. In the case of the third cranial nerve, the real origin is located in the mesencephalon tegmentum at the level of the superior colliculus in a nuclear complex: (1) the somatic nuclei for the extraocular muscles; and (2) the visceral nuclei (Edinger-Westphal nuclei) for parasympathetic pathways. From here, the fibers of the third cranial nerve continue ventrally until they leave the mesencephalon at the interpeduncular fossa, the site of apparent origin (Pierson & Carpenter, 1974). The nerve then follows a cisternal trajectory through the subarachnoid space, first in the interpeduncular cistern, in relation to the basilar artery branching into the posterior cerebral arteries, and passes between these and the superior cerebellar arteries.

It courses anterolaterally at the junction of the carotid, interpeduncular, prepontine, and cerebellopontine cisterns to reach the dural roof of the cavernous sinus at the oculomotor triangle. Before entering the wall of the cavernous sinus it is lateral to the internal carotid and posterior communicating arteries. It becomes incorporated into the fibrous lateral wall of the cavernous sinus, coursing superior to CN IV (Inoue et al., 2009; Rhoton, 2002a); then it leaves the intracranial space to enter the orbit at the medial part of the superior orbital fissure, with CNs IV and V1 and the ophthalmic vein. The nerve splits into superior and inferior divisions; the superior division enters the orbit below the attachment of the superior rectus muscle to reach the superior rectus and levator muscles; the inferior division reaches the medial and inferior rectus, the inferior oblique muscles; and the ciliary ganglion, which gives rise to the short ciliary nerves to the constrictor pupillae and ciliary muscles.

Eye movements are controlled by a complex system that comprise cortical, diencephalic, brain stem, and muscular structures that allow stable, precise, quick, smooth, and binocular eye movements. Six systems control the gaze to keep the fovea on target. These systems are the saccadic eye movements, that shift the fovea rapidly to a visual target in the periphery; smooth pursuit movements, that keep the image of a moving target on the fovea; vergence movements, that move eyes in opposite directions so that the image is positioned in the foveae; vestibulo-ocular movements, that hold images still on the retina during head movements; optokinetic movements that hold images during sustained head rotation driven by visual stimuli; and a fixation system that holds the eye still during intent gaze (Kandel, 2013). These movements are conducted by six extraocular muscles, four of which are innervated by

the third cranial nerve. The ocular motor signals of gaze are controlled by horizontal control centers in the brain stem by interneurons localized in the paramedian pontine reticular formation, vestibular nucleus, and nucleus prepositus hypoglossi, which are in functional relationship with third and sixth cranial nerve nuclei, and connected by the medial longitudinal fasciculus. Vertical movements are organized by interneurons localized in the rostral interstitial nucleus of the medial longitudinal fasciculus in the mesencephalic reticular formation. Both centers are controlled by higher cortical centers localized in the parietal and frontal cortices by the superior mesencephalic colliculus. The vermis and flocculus of the cerebellum also control eye movements related to vestibular reflexes and smooth pursuit. Vestibular nuclei control vestibulo-ocular reflexes that compensate for head movement, in functional relationship with cervical muscle control centers (Kandel, 2013; Ramat, Leigh, Zee, & Optican, 2007).

Anatomical Landmarks

- Orbit
- Superior orbital fissure
- Cavernous sinus
- Interpeduncular fossa
- Mesencephalon between substantia nigra and tectum

Figure 14.3 depicts the imaging anatomy of CN III.

Clinical Correlation

Lesions can affect the third nerve in the brain stem, subarachnoid space, and cavernous sinus at the superior orbital fissure or the orbit. Lesions affecting oculomotor nucleus cause ipsilateral third nerve palsy, with contralateral ptosis

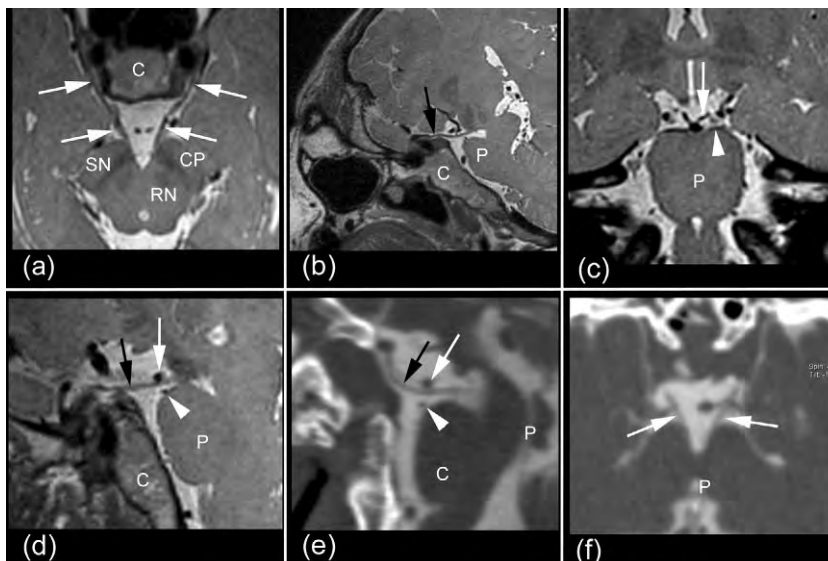


FIGURE 14.3 Oculomotor nerve. (a) Axial T₂-w MR at the level of the midbrain showing both CN III (white arrows) exiting anteriorly into the interpeduncular fossa. (b, d, e) Reconstructed sagittal T₂-w MR images and sagittal image from a CT cisternogram demonstrate CN III (black arrow) passing between the PCA (white arrow) and SCA (white arrowhead) as it exits the midbrain. (c) The same relationship can be seen in a coronal T₂-w MR image. (f) Reconstructed axial oblique image from a CT cisternogram showing CN III (white arrows) traversing the interpeduncular fossa outlined by contrast within the subarachnoid space. Note the cerebral peduncles (CP), substantia nigra (SN), red nuclei (RN), pons (P), and clivus (C).

and superior rectus paresis; lesions affecting the third nerve fascicle cause partial or complete isolated nerve palsy with or without pupil involvement. Lesions affecting the nerve in the cavernous sinus cause painful third nerve palsy with palsies of the fourth and sixth nerve with small pupil. Lesions in the superior orbital fissure can cause nerve palsy with or without palsies of the fourth and sixth nerve, often with proptosis. Lesions affecting the nerve in the orbit can affect superior or inferior branch with unilateral isolated palsies of extraocular muscles (Capo, Warren, & Kupersmith, 1992; Paul et al., 2007).

The most common complaint related to lesions in the ocular motor system is diplopia, which results more frequently with lesions of the extraocular muscles or ocular motor nerves (Brazis & Lee, 1998). Infarction, hemorrhage, tumors, infection, and demyelinating diseases cause nuclear and fascicular lesions; meningiomas, chordomas, metastases, infectious, or inflammatory processes of the meninges and trauma cause lesions in subarachnoid space. It can also be seen with transtentorial herniation (e.g., tumor, subdural or epidural hematoma) when an increased supratentorial pressure (e.g., from a tumor) forces the hippocampal uncus through the tentorial notch compressing or stretching the oculomotor nerve. The pupilloconstrictor fibers are affected first, resulting in a dilated, fixed pupil. The somatic efferent fibers are affected later, resulting in external strabismus (exotropia). Aneurysms of the carotid and posterior communicating arteries often compress CN III within the cavernous sinus or interpeduncular cistern. Diabetes mellitus (diabetic oculomotor palsy) can affect the oculomotor nerve by injuring the central fibers and sparing the pupilloconstrictor fibers (Fix, 2001). Cavernous sinus and superior orbital fissure can be affected by dural carotid-cavernous sinus fistula, cavernous sinus thrombosis or infection, superior ophthalmic vein thrombosis, tumors like pituitary adenomas and meningiomas, pituitary apoplexy, and mucocoele of the sphenoid sinus. The most common orbital lesions are infections, orbital tumors, and trauma. Denervation of the levator palpebrae muscle can cause ptosis, and denervation of the extraocular muscles causes the affected eye to look “down and out” as a result of the unopposed action of the lateral rectus and superior oblique muscles. Because the superior oblique and lateral rectus muscles are innervated by CN IV and CN VI, respectively, oculomotor nerve palsy results in diplopia when the patient looks in the direction of the paretic muscle. Interruption of the parasympathetic innervation (internal ophthalmoplegia) results in a dilated, fixed pupil and paralysis of accommodation (cycloplegia).

The third cranial nerve can be approached by pterional approach and its variants, which include the cavernous sinus approaches, for decompression of the nerve in tumoral and vascular lesions. It can also be reached by the subtemporal route.

Trochlear Optic Nerve

Function

This nerve controls the superior oblique muscle contraction, which rotates the eyeball internally in abduction and lowers the eyeball in adduction.

Anatomy

CN IV has its real origin in the midbrain tegmentum at the level of the inferior colliculus, at the fourth nerve motor nucleus. The radicular fibers then travel in a dorsolateral direction with the apparent origin in the dorsal aspect of the midbrain, just below the inferior colliculus. The nerve travels around the midbrain between the superior cerebellar and the posterior cerebral arteries, in the lateral mesencephalic cisterns, in lateral relation to the tentorial incisura (Inoue et al., 2009). It then enters the roof of the cavernous sinus posterolateral to the third nerve and courses below the oculomotor nerve in the posterior part of the lateral wall; anteriorly, it passes upward along the lateral surface of the oculomotor nerve. From there, the trochlear nerve passes medially between the third nerve and the dura lining the lower margin of the anterior clinoid and optic strut to reach the medial part of the orbit and the superior oblique muscle through the superior orbital fissure (Rhoton, 2002a).

Anatomical Landmarks

- Orbit
- Superior orbital fissure
- Tentorial incisura
- Inferior midbrain colliculus

Figure 14.4 shows the cisternal segment of CN IV.

Clinical Correlation

CN IV paralysis can lead to the following conditions: extorsion of the eye and weakness of downward gaze; vertical diplopia, which increases when looking down; and head tilting to compensate for extorsion (potentially misdiagnosed as idiopathic torticollis). Etiologies for a fourth nerve palsy localized in the midbrain are more frequently due to demyelinating diseases, ischemia, infarction, hemorrhage, and tumors. Subarachnoid lesions are due to aneurysms of the superior cerebellar artery, hydrocephalus, infections, neoplasm-like meningiomas, metastasis, schwannoma, pineal tumors, and trochlear nerve sheath tumors. Head trauma can be a cause. Because the trochlear decussation underlies the superior medullary velum, a trauma at this site often results in bilateral fourth nerve palsies. Also, pressure against the free border of the tentorium (herniation) can injure the nerve (Fix, 2001). Cavernous sinus lesions that can cause nerve palsy are meningiomas, pituitary tumors, inflammation, internal

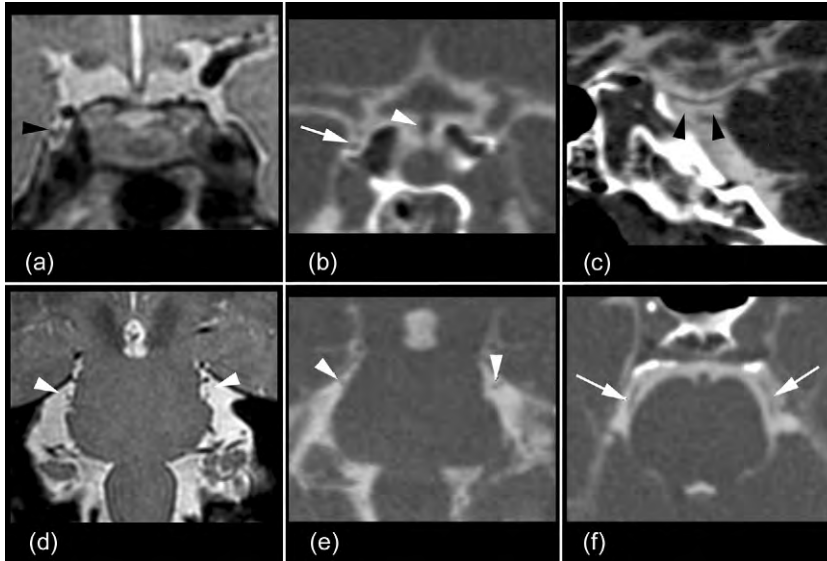


FIGURE 14.4 Trochlear nerve. (a) Coronal T₂-w MR image showing heterogeneous signal intensity involving the right cavernous sinus (*white arrowhead*). (b) Coronal CT cisternogram image at the level of the pituitary stalk (*white arrowhead*) demonstrating the locations of CN III and IV in the lateral wall of the cavernous sinus (*white arrow*). (c) Sagittal image from a CT cisternogram depicts the cisternal segment of CN IV (*black arrowheads*) running anteriorly and traversing the pontomesencephalic cistern. (d) Coronal T₂-w MR and (e) coronal CT cisternogram images present a ventral view of the brain stem depicting the course of CN IV (*white arrowheads*) through the ambient (perimesencephalic) and crural cisterns. (f) Axial CT cisternogram image at the level of the inferior colliculus of the midbrain reveals the course of CN IV (*white arrows*).

carotid aneurysm, and dural carotid and cavernous sinus fistula. Orbit lesions related to fourth nerve palsy are similar to those mentioned previously with third nerve palsy.

Trigeminal Nerve

Function

The fifth cranial nerve transmits sensations from the skin of the anterior part of the head, the nasal and oral cavities, teeth, and meninges. It also carries motor fibers to the masticatory muscles.

Anatomy

The motor fibers have their origin in the pons, in a medial and dorsal position; the sensitive fibers originate in the trigeminal ganglia. The trigeminal nerve has its apparent origin at the junction of the ventral surface of the pons and the middle cerebellar peduncle. It is the coarsest cranial nerve and travels across the subarachnoid space at the cerebellopontine cistern in a rostral and lateral direction, reaching the trigeminal ganglia at the ventral surface of the petrous part of the temporal bone. Here, it is related to the caudal loop of the superior cerebellar artery (Revuelta-Gutierrez, Martinez-Anda, Coll, Campos-Romo, & Perez-Pena, 2013). At this point, it splits into three main divisions. First, the ophthalmic nerve (V1), the smallest of the three trigeminal divisions, which is inclined upward as it passes forward near the medial surface of the dura, forms the lower part of the lateral wall of the cavernous sinus, to reach the superior orbital fissure. It reaches the superior orbital fissure and splits into the lacrimal,

frontal, and nasociliary nerves. The lacrimal nerve courses along the superior margin of the lateral rectus muscle, where it receives secretory fibers from the pterygopalatine ganglion and is distributed to the lacrimal gland. The frontal nerve passes on the medial side of the origin of the lateral rectus muscle and divides into the supratrochlear and supra-orbital nerves within the orbit, conveying sensation from the upper eyelid and forehead. The nasociliary nerve courses medially with the abducens nerve and carries sensation from the cornea, eyeball, and sclera; it also conveys sympathetic fibers to the globe and pupillary dilator (Rhoton, 2002c). Second, the maxillary division enters the lateral wall of the cavernous sinus at the anteromedial middle fossa triangle until it reaches the foramen rotundum (Rhoton, 2002a). Then it reaches the pterygopalatine fossa, where it is situated superior to the maxillary artery. Here, it gives rise to six collateral rami: the middle meningeal, zygomatic, pterygopalatine, superoposterior, middle superior, and superoanterior alveolar nerves. Its terminal branch is the infraorbital nerve, which reaches the orbit through the inferior orbital fissure, traveling across the infraorbital conduct to reach the infraorbital orifice. It conducts sensation from the cheek, inferior lip, nasal wing, and superior lip. It also conducts sensation from the nasal mucosa, maxillary teeth and gum, and the parietal and temporal dura, and it conducts parasympathetic fibers for nasal secretion. Third, the maxillary nerve marks the inferior limit of the anterolateral middle fossa triangle of the cavernous sinus and reaches the foramen rotundum, passing across it to reach the infratemporal fossa, where it divides in two terminal rami, the anterior and posterior trunks. Its main terminal ramus is the inferior alveolar nerve, which travels across the mandible to reach the mandibular orifice.

This division conducts sensation from the temporal region, cheek, chin, oral cavity, inferior lip, anterior tongue, lower teeth, and mandible, and it also conducts sensation from the middle fossa dura. Its motor division innervates the masticatory muscles (Agur & Dalley, 2013).

Anatomical Landmarks

- Pons
- Middle cerebellar peduncle
- Cerebellopontine angle
- Petrous bone
- Great wing sphenoid
- Orbit
- Infratemporal fossa

Figure 14.5 shows the anatomical landmarks of CN V.

Clinical Correlation

The fifth cranial nerve has an extensive distribution area and may be affected by a variety of pathologies. Its clinical evaluation consists of sensory evaluation of the cutaneous area covered by roots of the trigeminal nerve from the cranial vertex to the mandible. It also includes evaluation of the motor functions of masticatory muscles, and the corneal, masseter, blink, and corneomandibular reflexes.

Lesions of CN V result in at least six different neurological deficits:

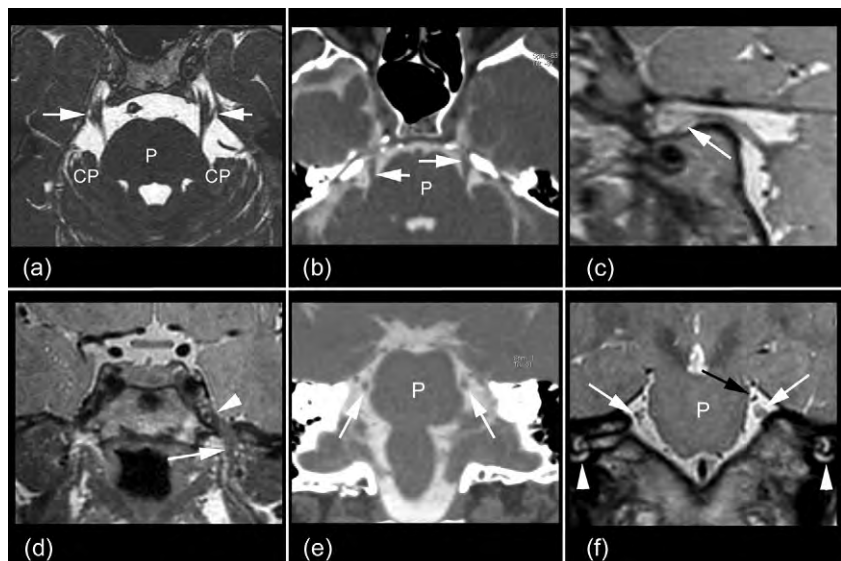
1. Loss of general sensation (hemianesthesia) from the face and mucous membranes of the oral and nasal cavities.
2. Loss of the corneal reflex (afferent limb, CN V₁).
3. Flaccid paralysis of the muscles of mastication.

4. Deviation of the jaw to the weak side as a result of the unopposed action of the opposite lateral pterygoid muscle.
5. Paralysis of the tensor tympani muscle, which leads to hypoacusis (partial deafness to low-pitched sounds).
6. Trigeminal neuralgia (tic douloureux), which is characterized by recurrent paroxysms of sharp, stabbing pain in one or more branches of the nerve (Fix, 2001).

Lesions affecting the trigeminal nerve can be localized in the brain stem, subarachnoid space, trigeminal ganglion, and petrous apex. It can also be affected after it gives rise to its three roots in the cavernous sinus and outlet orifices. Motor and sensory nuclei of the trigeminal nerve may be involved by lesions that affect the pons, medulla, or upper cervical cord, and these lesions are part of more complex syndromes affecting adjacent long tracts and other cranial nerves. Most frequent lesions in this localization are demyelinating diseases, infarction, hemorrhage, brain stem gliomas, and arteriovenous malformations (Bathla & Hegde, 2013).

In its cisternal course, the preganglionic trigeminal nerve root may be damaged by tumors, such as meningiomas, schwannomas, and metastasis; and by infection, aneurysm, trauma, and vascular contact of cerebellar arteries (trigeminal neuralgia). This localization is suggested by the involvement of the neighboring cranial nerves. Lesions affecting trigeminal ganglion are those localized in the middle cranial fossa and petrous apex, the most frequent being tumors (trigeminal schwannoma, meningiomas, metastasis, and chordomas), inflammatory diseases, infections, trauma, complicated otitis media, and abscesses (Bathla & Hegde, 2013; Becker, Kohler, Vargas, Viallon, & Delavelle, 2008). Primary trigeminal

FIGURE 14.5 Trigeminal nerve. Axial (a) T₂-w MR and (b) CT cisternogram images illustrating the course of CN V (white arrows) exiting the mid-lateral pons (P) at the level of the middle cerebellar peduncles (CP). (c) Reconstructed sagittal CT cisternogram demonstrates CN V (white arrow) dividing into smaller branches within Meckel's cave (white arrow). (d) Coronal T₂-w MR image, at the level of the midsphenoid sinus, shows Meckel's cave (white arrowhead) and CN V₃ as it exits through the foramen ovale (white arrow). (e) Coronal T₂-w MR and (f) CT cisternogram images reveal the cisternal segments of CN V (white arrows) at the lateral aspect of the midpons (P) and the relationship between the dorsal surface of the nerve and the superior cerebellar artery (black arrow); both cochleas in inferolateral position are easily visible in the last image (white arrowhead).



neuralgia is thought to be caused by vascular compression on the fifth cranial nerve by a cerebellar artery. The pulsatile compression causes demyelination of the trigeminal nerve, and triggers autonomous discharges that cause the characteristic electric shock-like pain of this disease. This pathology is successfully treated by microvascular decompression of the nerve, a procedure that consists of separating the nerve from the offending artery by microsurgical technique (Revuelta-Gutierrez et al., 2013).

Lesions affecting the cavernous sinus can damage the first and second roots of the trigeminal nerve, and these lesions usually affect the third, fourth, and sixth cranial nerves. Lesions affecting the peripheral branches of the trigeminal nerve are localized in the superior orbital fissure and orbit, in the case of V1; and the middle cranial and infratemporal, pterygopalatine fossa, and face in the cases of V2 and V3. Most frequent lesions in this localization are craniofacial tumors and trauma (Becker et al., 2008).

Surgical approaches to the trigeminal nerve on its posterior fossa path are done by retrosigmoid procedures, including the keyhole asterional approach for trigeminal microvascular decompression. The subtemporal, pretemporal, cavernous sinus, and laterally extended transsphenoidal endoscopic routes can reach trigeminal ganglion. Final branches of the trigeminal nerve require different approaches. Ophthalmic division can be approached by the pterional route; maxillary and mandibular rami can be reached by infratemporal procedures in craniofacial surgery.

Abducens Nerve

Function

This nerve supplies the lateral rectus muscle, which abducts the eyeball.

Anatomy

The sixth nerve originates in the abducens motor nucleus, which is located at the dorsal aspect of the pons near the floor of the fourth ventricle, surrounded by the motor fibers of the seventh nerve. The radicular fibers then travel ventrally, arriving at the pontomedullary junction near the midline. It travels between the prepontine and cerebellopontine cisterns, having the longest subarachnoid course of all CNs. It pierces the dura forming the lower part of the posterior wall of the sinus at the upper border of the petrous apex and enters Dorello's canal (Ezer, Banerjee, Thakur, & Nanda, 2012), where it passes below the petrosphenoid ligament to enter the cavernous sinus at the level of the middle clivus. The nerve bends laterally around the proximal portion of the intracavernous internal carotid artery and ascends as it passes inside the cavernous sinus medial to the ophthalmic nerve. It has the most medial site

of entry of the nerves coursing through the sinus. It passes through the superior orbital fissure and annular tendon to enter the medial surface of the lateral rectus muscle (Rhoton, 2002a, 2002c).

Horizontal gaze is controlled by premotor burst neurons localized in the paramedian pontine reticular formation below and extending rostrally from the level of the abducens nucleus and projecting monosynaptically to the ipsilateral abducens nucleus, and to a network of neurons in the nucleus prepositus hypoglossi and adjacent medial vestibular nucleus that contribute to the final common integrator's horizontal component. They send their axons across the midline to the opposite middle longitudinal fasciculus, where they ascend through the pons middle longitudinal fasciculus where they ascend through the pons and midbrain to end in the third nerve nucleus, coordinating the action of both eyes to produce horizontal gaze. Frontal eye field controls the horizontal movement of both eyes through its connection with superior mesencephalic colliculus (Schall, 2013).

Anatomical Landmarks

- Fourth ventricle
- Middle clivus
- Cavernous sinus and cavernous segment of internal carotid artery
- Superior orbital fissure

Figure 14.6 depicts the imaging landmarks of CN VI.

Clinical Correlation

CN VI paralysis is the most common isolated palsy and results from the long peripheral course of the nerve. An abducens nerve paralysis can result in the following defects: convergent (medial) strabismus (esotropia), with the inability to abduct the eye; and horizontal diplopia with maximum separation of double images when looking toward the paretic lateral rectus muscle (Fix, 2001). Lesions affecting the abducens nucleus cause ipsilateral gaze palsy to the same side. Lesions affecting dorsolateral pons and abducens fascicle cause ipsilateral gaze palsy, facial paresis, and (occasionally) contralateral hemiparesis. The most common lesions localized here are demyelinating lesions, ischemia or infarction, hemorrhage, and brain stem gliomas. Lesions to the abducens nerve cause ipsilateral lateral rectus muscle palsy with affection of the facial nerve (petrous apex), third and fourth nerve palsies (cavernous sinus), and proptosis (superior orbital fissure syndrome) (Dekeyser & Lemmerling, 2009).

Surgical approaches to the abducens nerve include retrosigmoid, extended posterior and anterior transpetrosal, cavernous sinus, and transnasal-transclival approaches.

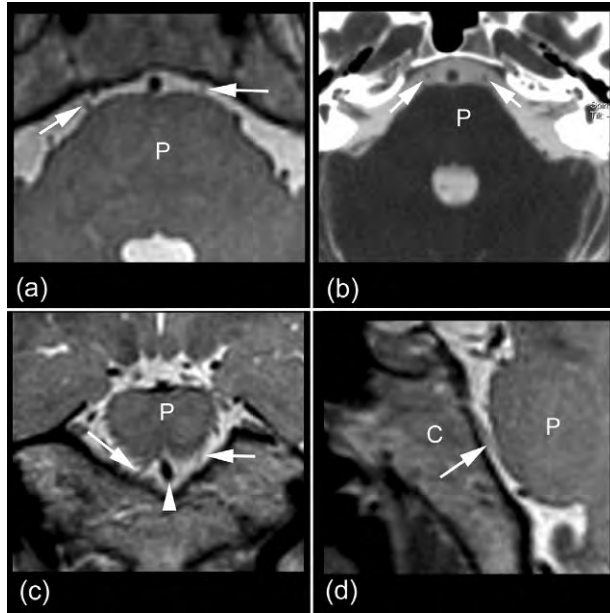


FIGURE 14.6 Abducens nerve. (a) Axial T₂-w MR and (b) CT cisternogram images reveal the bilateral cisternal segment of CN VI (white arrows) exiting the ventral aspect of the pontomedullary junction. (c) Reconstructed coronal oblique T₂-w MR image depicts the ascending course of CN VI in the prepontine cistern lateral to the basilar artery (white arrowheads). (d) Sagittal T₂-w MR image demonstrates the long cisternal course of CN VI (white arrow) in fine detail as it courses toward Dorello's canal; here we see its relationship with the pons (P) and clivus (C). CN VI is the longest CN.

At its entrance to the orbit, it can be reached by a pterional approach.

Facial Nerve

Function

The main function of the facial nerve is to supply the muscles of facial expression. Its other functions are to conduct taste sensation from the anterior tongue and oral cavity and to effect parasympathetic secretion from the salivary, lacrimal, nasal, and palatine glands (Zhang et al., 2013).

Anatomy

The facial nerve has three origins: (1) the motor nuclei, in the dorsal part of the pons; (2) the superior salivary nuclei, in the dorsolateral portion of the pons; and (3) the geniculate ganglion, which receives taste sensation from the anterior tongue. It leaves the brain stem at the cerebellopontine angle (at the lateral part of the pontomedullary junction) as two separate nerves, the intermediate and facial nerves.

It has five segments (Binder et al., 2010): (1) the intracranial segment; (2) the internal auditory canal segment; (3) the labyrinthine segment; (4) the tympanic segment; and (5) the mastoid segment. The intracranial segment passes through the subarachnoid space at the cerebellopontine

cistern, medial to CN VIII and inferolateral to CN V, in a mediolateral direction. The anteroinferior cerebellar artery (AICA) arises at the pontine level and courses in relation to the facial and vestibulocochlear nerves to reach the surface of the middle cerebellar peduncle. It reaches the internal acoustic meatus, which runs anterior to the superior vestibular nerve and superior to the cochlear nerve (Rhoton, 2000b). At the internal auditory canal segment, the facial nerve maintains its superior position relative to the cochlear nerve and passes above the transverse crest to enter the fallopian canal. The labyrinthine segment heads anteriorly and laterally, running superior to the cochlea and vestibule, until it reaches the geniculate ganglion. Distal to the ganglion, the facial nerve turns abruptly posterior to form the first genu of the facial nerve, which reaches the fallopian canal. The tympanic segment parallels the longitudinal axis of the petrous pyramid, running posteriorly and laterally on the medial wall of the tympanic cavity, between the lateral canal superiorly and the oval window inferiorly. At the sinus tympani, the nerve turns inferiorly to form the second genu. The mastoid segment goes vertically downward in the posterior wall of the tympanic cavity and the anterior wall of the mastoid, to its egress from the skull through the stylomastoid foramen. On this trajectory, the nerve gives rise to the greater petrosal nerve, chorda tympani, and the nerve to the stapedius. It then runs inferiorly, reaching the base of the styloid process, crossing the parotid gland, passing between the digastric and styloid muscles. In the parotid, the nerve runs anteriorly, inferiorly, and laterally, reaching the external jugular vein, where it divides in two main trunks, which yield five terminal rami: the temporal, cigomatic, oral, mandibular, and cervical (Monkhouse, 2006).

Motor function represents the final pathway of mimic movement control. It is controlled by higher centers localized in the frontal brain lobe. All social intercourse is colored by facial expressions, and these may be intentional, but more often they are produced without conscious control. Facial expressions are produced by synergistic or cooperative action of many different facial muscles. One mechanism that could account for such muscle synergy would be that it is produced by the branching pattern of common stem, last order, presynaptic input fibers innervating motoneurons applying the different muscles, and in this way a hardwired repertoire of muscle synergies for the different expressions is encoded (Root & Stephens, 2003). The limbic system also plays a fundamental role in somatic expression of emotion, and considerable evidence from both humans and experimental animals indicates that the amygdala and hippocampus intervenes between the regions concerned with the somatic expression of emotion, that is, the hypothalamus and brain stem nuclei, and the neocortical areas concerned with conscious feeling, that is, the cingulate, parahippocampal, and prefrontal cortices.

Taste is detected by taste cells localized in taste buds on the tongue, palate, pharynx, epiglottis, and upper third of the esophagus. Each taste bud has a taste pore where microvilli extends to make contact with molecules that produce taste in taste cells. Those cells are innervated by sensory neurons at its basal pole. The gustatory cells distinguish four basic qualities of stimulus: bitter, salty, sour, and sweet, and the combination of these basic tastes give us all taste sensations. The taste buds in the anterior two-thirds of the tongue are innervated by sensory neurons of the geniculate ganglion, whose peripheral branches travel in the chorda tympani, a branch of the seventh cranial nerve. Those taste buds localized in the posterior third of the tongue are innervated by the ninth cranial nerve, and those localized in the epiglottis and the esophagus travel by the tenth cranial nerve. The three inputs enter the solitary tract in the medulla, and synapse on the column of cells in the gustatory area of the rostral part of the nucleus of the solitary tract. These neurons project to the thalamus, where they terminate in the parvocellular region of the ventral posterior medial nucleus. These neurons project along the border between the anterior insula and the frontal operculum in the ipsilateral cortex (Kandel, 2013).

Anatomical Landmarks

- Middle cerebellar peduncle
- Internal auditory canal
- Anteroinferior cerebellar artery

- Cochlea and semicircular canals
- Tympanic cavity
- Mastoid
- Styloid process
- Parotid gland

Figure 14.7 demonstrates the main imaging landmarks of CN VII.

Clinical Correlation

Clinical evaluation of CN VII includes motor, sensory, and autonomous functions. Motor functions of the innervated facial muscles are assessed by facial inspection and tests of facial mobility. The symmetry of blinking and lip movements are noted, and the patient may be asked to raise the eyebrows, wrinkle the brow, close the eyes, show the teeth, blow the cheeks, and retract the chin. One useful scale is the House-Brackmann facial palsy grading system (House & Brackmann, 1985). Sensory examination of the facial nerve consists of the evaluation of taste on the anterior two-thirds of the tongue with the four fundamental tastes mentioned earlier. The most important reflexes supplied by the facial nerve are corneal and palpebral reflexes. For parasympathetic function Schirmer's test can be used, putting a filter paper on each lower lid for lacrimal evaluation (Paul et al., 2007).

Lesions to CN VII cause the following conditions:

1. Flaccid paralysis of the muscles of facial expression (upper and lower face).

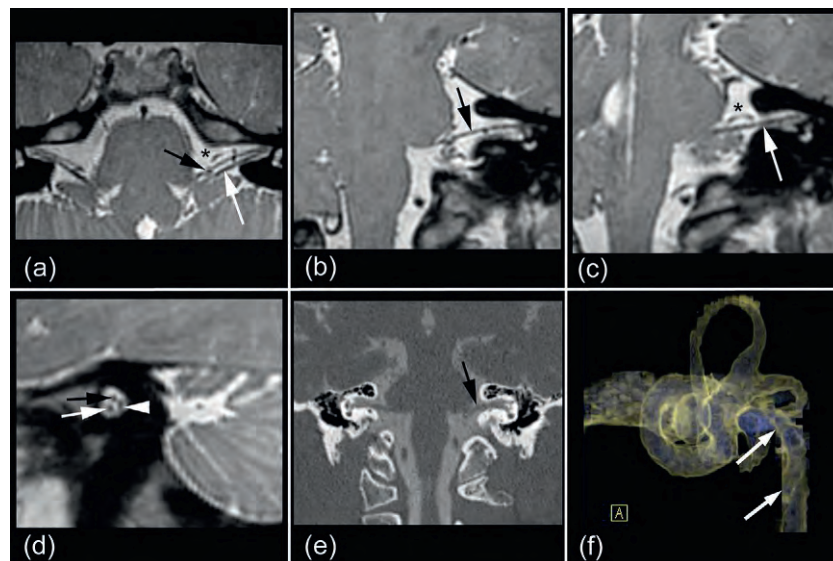


FIGURE 14.7 Facial and vestibulocochlear nerves. (a) Reconstructed axial oblique T₂-w MR image through posterior fossa illustrating CN VII (black arrow) and VIII (white arrow) exiting the lateral aspect of the pontomedullary junction and coursing into the internal auditory canal (IAC); in this image we see the upper medulla (M) and the belly of the pons (P), which gives the angle of slice acquisition. (b, c, e) Coronal T₂-w MR and CT cisternogram images depicting normal IAC with CN VII (black arrow) and VIII (white arrow) coursing through them; in (a) and (c) the close relationships of both nerves to the anterior inferior cerebellar artery (black asterisk) are noticeable. (d) Oblique parasagittal T₂-w MR image through the IAC reveals the normal positions of CNs VII and VIII; superomedially located, CN VII (black arrow) is slightly smaller than the inferomedially located cochlear nerve (white arrow); laterally located, the vestibular component of CN VIII (white arrowhead) is starting to divide into its superior and inferior divisions. (f) Oblique coronal CT cisternogram image depicts some segments of the CN VII through the right temporal bone: the tympanic (black arrowhead), also called the horizontal segment, and the mastoid (black arrow), also called the vertical segment.

2. Loss of the corneal reflex (efferent limb), which can lead to corneal ulceration.
3. Loss of taste (ageusia) from the anterior two-thirds of the tongue, which can result from injury to the chorda tympani.
4. Hyperacusis (increased acuity to sounds) as a result of stapedius paralysis.

The localization of lesions affecting the facial nerve is based on the anatomy of the facial nerve pathway and branches. In supranuclear corticobulbar lesions there is contralateral paresis of the lower portion of the face with sparing of upper facial function because of bilateral supranuclear control. Nuclear and fascicular lesions are identified because they usually affect neighboring structure, such as the abducens nucleus, paramedian pontine reticular formation, and spinal tract. Supranuclear and nuclear, the most frequent lesions are vascular, demyelinating, and tumoral in origin.

Posterior fossa lesions localized in the cerebellopontine angle result in peripheral facial nerve lesion on its sub-arachnoid portion causing ipsilateral paralysis, including all facial functions, and ipsilateral tinnitus, deafness, and vertigo. Lesions on this location are tumoral, the most frequent tumors being acoustic neuroma, meningioma, and epidermoid cyst. The nerve can be affected in the meatal canal with similar findings in patients with skull base fractures. Within the facial canal, the nerve can be affected distal from geniculate ganglion but proximal from departure of the nerve to the stapedius muscle with all facial nerve functions affected, except that the patient has lacrimal function intact. If the lesion is distal to the stapedius muscle, but proximal to the chorda tympani, the patient also preserves the stapedius reflex and has no hyperacusis. If the lesion is distal to the chorda tympani origin, the patient also preserves taste. Lesions on the facial canal can be related to varicella zoster infection (Ramsay Hunt syndrome), complicated otitis media, cholesteatoma, glomus jugulare tumors, osteomyelitis, and carcinoma of the ear.

Surgical approaches to facial nerve include retrosigmoid and posterior transpetrosal approaches; and head-neck surgery procedures for its extracranial path from its emergence through the stylomastoid foramen to its final branches.

Vestibulocochlear Nerve

Function

The vestibulocochlear nerve is the sensory nerve for hearing (cochlear) and equilibration (vestibular).

Anatomy

Hearing depends on transduction of mechanical stimuli to elicit an electrical response in hairy cells in the organ of Corti, which respond to specific stimulus frequencies. This is

possible because of the amplification of sound waves by the tympanic membrane and middle ear bones, and by the oval window and basilar membrane on the inner ear. The organ of Corti has hair cells with villi that permits transduction of mechanical stimuli to electrical activity, which is then conducted by spiral ganglion neurons to the brain stem. Vestibular input originates from hair cells stimulation of gelatinous material embedded with otoconias (calcium carbonate stones) on the vestibular labyrinth. This organ can measure linear and angular acceleration. Linear acceleration is detected by the utricle and the saccle; angular accelerations are measured by the semicircular canals. These stimuli are conducted by scarpa ganglion neurons to originate the vestibular division of the eighth nerve. From here, central processes pass the cochlear nerve to the cochlear and vestibular nuclei (Monkhouse, 2006). CN VIII passes across the internal auditory canal and divides into three branches: the cochlear nerve, which passes medially and inferiorly; the superior vestibular nerve, which passes laterally and superiorly; and the inferior vestibular nerve, which passes inferiorly and laterally to the vertical and horizontal crests of the IAC (Agur & Dalley, 2013). The nerve then passes to the subarachnoid space at the cerebellopontine cistern, lateral to the facial nerve, and in relation to the meatal and postmeatal segment of the AICA. It runs in a medial and posterior direction to reach the cerebellopontine angle of the brain stem, crossing the pontomedullary sulcus (Rhoton, 2000a). The cochlear nerve reaches the cochlear nuclei laterally in the floor of the fourth ventricle, the sensory neurons of which form the lateral lemniscus tract to inferior colliculi, dorsally in the midbrain. From here, the tract continues to the medial geniculate bodies in the posterior and medial part of the thalamus. These nuclei form the auditory radiation fibers in the sublenticular section of the internal capsule, reaching Heschl's gyri at the opercular surface of the temporal lobe (Wen et al., 1999).

The vestibular nerve reaches the vestibular nuclei in the floor of the fourth ventricle; vestibular impulses pass to the floccular nodule, uvula, and fastigial nucleus of the cerebellum, crossing through the inferior cerebellar peduncle. The vestibular nuclei also project to the spinal cord (vestibulospinal tract), the eye muscle nuclei (medial longitudinal fasciculus), and the ventral posterior nucleus of the thalamus. The vestibular system originates reflexes that stabilize the eyes and body when the head moves; and vestibulo-ocular reflexes that compensate orbits position for head movements.

Anatomical Landmarks

- Cochlea and vestibule
- Inner ear
- IAC
- Cerebellopontine angle

- AICA
- Lateral part of floor of fourth ventricle

Figure 14.7 shows the cisternal landmarks of CN VIII.

Clinical Correlation

CN VIII-related symptoms are deafness, autophony, tinnitus, vertigo, imbalance, and nystagmus. Cochlear branch function is evaluated to define if hearing loss is sensorineural, conductive (between the environment and the organ of Corti), or mixed. Assessment begins with inspection of the external auditory canal and tympanic membrane by otoscopy. Then, qualitative assessment of hearing loss is done by a tuning fork of 512 Hz using three major tuning fork tests: Weber's test, which helps differentiate a conductive from a sensorineural hearing loss in a unilateral hearing loss; Rinne's test, which compares the patient air and bone conduction; and Schwabach's test, which compares examiner bone conduction to that of the patient (Paul et al., 2007).

Hearing loss evaluation continues with audiological tests. These tests include pure tone audiometry, speech recognition, immittance battery, otoacoustic emissions, electrophysiology (e.g., auditory brain stem tests), middle latency responses, auditory steady state response, electrocochleography, and evoked cortical potentials.

Vertigo is the most common complaint of vestibular dysfunction. Maintaining balance is a complex task requiring sensory inputs from the vestibular, visual, and somatosensory systems, which are integrated at the vestibular nuclei and entail two major reflexes, the vestibulo-ocular reflex and the vestibulospinal reflex. The task of making a satisfactory diagnosis of the cause of vertigo is daunting; in most cases, laboratory tests and radiological investigations are not helpful in making a diagnosis. The causes of vertigo are most commonly ontological. The first question is to determine if the patient complaint is vertigo, which is a hallucination of movement, and it is typically but not necessarily rotatory. Next it has to be determined if vertigo is central or peripheral in origin; semicircular canals, saccule, utricle, and vestibular nerve constitute the peripheral vestibular system; vestibular nuclei, cerebellum, brain stem, spinal cord, and vestibular cortex made up the central vestibular system. Peripheral vertigo is usually acute in onset, suppressed by visual fixation, and has severe associated symptoms of nausea and vomiting. There are otological concomitant symptoms, mild to moderate instability, and short duration, tending to decrease after a few days. Nystagmus is horizontal or torsional, with the same direction in all gazes, and fatigable. Central vertigo is subacute or slow in onset, not suppressed by visual fixation, with concomitant neurological rather than ontological symptoms, has severe associated instability, and persists after its onset with little or no improvement. Nystagmus associated with central vertigo is not fatigable and can be vertical or other direction, and may change direction with a change in gaze (Lee, 2012).

Testing and evaluation of a patient with vertigo begins with a complete ear, nose, and throat exam. Oculomotor function tests are performed by asking the patient to gaze at the tip of the clinician's index finger. In assessing for the presence of nystagmus, the clinician should be aware of spontaneous nystagmus, and the direction of its fast phase, frequency, and amplitude should be noted. If no spontaneous nystagmus is noted, it can be elicited by gaze and head shaking; and positional tests, like the Dix-Hallpike maneuver, are performed to elicit typical nystagmus of benign paroxysmal positional vertigo. Postural control tests include the Romberg, past-pointing, tandem gait, and the Fukuda stepping tests. Electronystagmography is the first step in a vestibular testing battery to evaluate the vestibulo-ocular reflex. Oculomotor, saccade, gaze, fixation suppression, positional, caloric, rotatory chair, and smooth pursuit testing can also be done. Vestibular evoked myogenic potentials are short latency electromyograms that provide diagnostic information about saccular and inferior vestibular nerve function (Della-Morte & Rundek, 2012).

Human auditory cortex lesions located in the superior temporal gyrus along the sylvian fissure do not cause complete deafness, even with bilateral, irritative lesions that may result in subjective auditory hallucinations. Brain stem lesions involving the auditory pathways do not cause hearing impairment. The most common lesions in the central nervous system are hemorrhage, infarction demyelinating, and brain stem gliomas.

Peripheral nerve lesions located in the cerebellopontine angle account for partial or complete deafness, often associated with ipsilateral tinnitus; this is secondary to trauma, infections, drugs (macrolides), aneurysms of the anterior inferior cerebellar artery, or tumors (vestibular schwannomas, meningiomas, and epidermoid cysts). There may be dysfunction of neighboring cranial nerves. Peripheral causes of vertigo are due to labyrinthine disease and nerve inflammation. Benign paroxysmal positioning vertigo is a very common mechanical disorder of the inner ear in which there are brief attacks of acute and severe vertigo with concomitant nystagmus and autonomic symptoms, usually self-limited. Other causes of positional vertigo include trauma, infection, ischemia, demyelinating disease, chiari malformations, posterior fossa tumors and perilymphatic fistulas, Meniere's disease, acute vestibular neuronitis, and acute labyrinthitis.

Surgical approaches to the vestibulocochlear nerve include the retrosigmoid and posterior transpetrosal routes.

Glossopharyngeal Nerve

Function

The main function of the glossopharyngeal nerve is in the gag reflex, because it provides the sensory supply from the oropharynx and posterior part of the tongue. Its other

functions are motor supply to the stylopharyngeus and the superior constrictor of the pharynx, conveying parasympathetic fibers part of the way to the parotid gland, and sensory supply from the carotid sinus, carotid body, and skin of the external acoustic meatus and tympanic membrane (Singh & Husain, 2011).

Anatomy

The glossopharyngeal nerve is closely related to the vagus nerve, with which it shares the bulbar nucleus. Its somatic sensory component originates in the superior ganglion, and the visceral sensory fibers originate in the inferior ganglion, both situated at the jugular foramen. The efferent component originates in the inferior salivary (parasympathetic fibers) and ambiguus nucleus (Branchiomotor fibers), both located within the medulla (Burchiel, 2011).

Fibers pass through medulla as one or, rarely, two rootlets along the posterior edge of the olive in the postolivary sulcus, above and in series with those of CNs X and XI. It then courses ventrally to the choroid plexus protruding from the foramen of Luschka, to the subarachnoid space in the cerebellomedullary cistern. Here it is in relation to the posteroinferior cerebellar artery or the vertebral artery; both can distort the rootlets of the nerve (Rhoton, 2000b). It courses inferolaterally to reach the jugular foramen in the groove leading from the pyramidal fossa below the opening of the cochlear aqueduct and along the medial side of the intrajugular ridge. After the nerve exits the jugular foramen, it turns forward, crossing the lateral surface of the internal carotid artery deep below the styloid process. As the nerve transverses the jugular foramen, it expands at the site of its superior and inferior ganglia (Rhoton, 2000c). Inferiorly to the cranial cavity it is situated posterior to the internal carotid, crossing it laterally, leaving

CNs X and XI posteriorly. It continues obliquely anteriorly and inferiorly, crossing the stylopharyngeus muscle, and over the superior constrictor muscle (Agur & Dalley, 2013).

Anatomical Landmarks

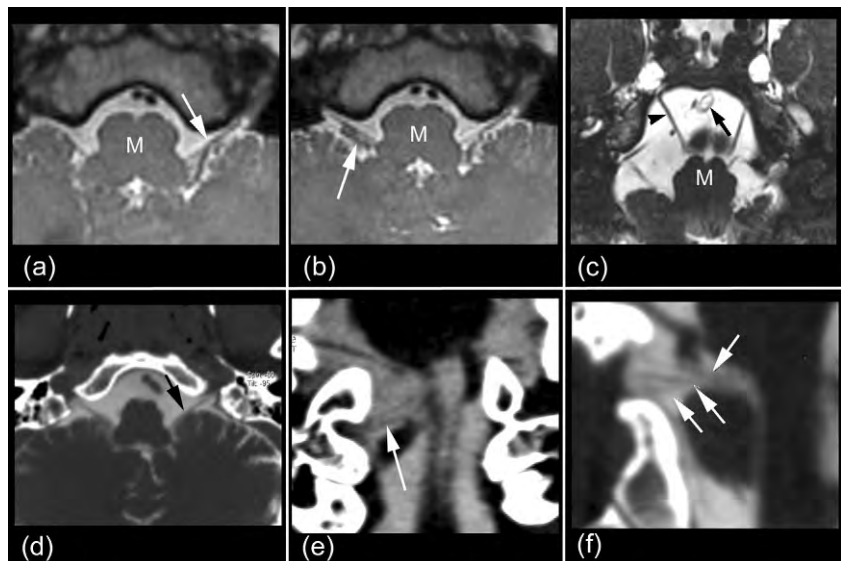
- Fourth ventricle floor
- Medullary olive
- Jugular foramen
- Internal carotid artery
- Pharynx

Figure 14.8a and d shows the cisternal landmarks of CN IX.

Clinical Correlation

Stylopharyngeal function is difficult to assess, Motor paresis may be negligible with glossopharyngeal nerve lesions. Sensation is tested on the soft palate, the posterior third of the tongue, the tonsillary regions, and the pharyngeal wall. The pharyngeal or gag reflex is tested by stimulating the posterior pharyngeal wall, tonsillar area, or base of the tongue. Salivary secretion may be decreased (Fix, 2001). Supranuclear, nuclear, and intramedullary lesions include pseudobulbar syndromes, syringobulbia, demyelinating disease, vascular disease, motor neuron disease, and intraaxial tumors. Subarachnoid space tumors are due to cerebellopontine angle lesions; glossopharyngeal neuralgia is due to vascular contact by posterior inferior and anterior inferior cerebellar arteries and is characterized by electric shock-like pain located in the tongue base and the posterior wall of the pharynx. Jugular foramen syndrome is uncommon and due to glomus jugulare tumors, basal skull fractures, neuroma, metastasis, cholesteatoma, meningioma, and infection. Lesions within retropharyngeal and retroparotid space (neoplasms, abscesses, adenopathy,

FIGURE 14.8 Glossopharyngeal, vagus, spinal accessory, and hypoglossal nerves. (a, d) Axial T₂-w MR and CT cisternogram images, depicting the cisternal portion of CN IX (white and black arrows) exiting the dorsolateral medulla (M) to traverse the more anterior part of the jugular foramen (*pars nervosa*). (b, e) Axial T₂-w MR and reconstructed coronal CT cisternogram images illustrating CN X and XI (white arrows) passing through the posterior part of the jugular foramen (*pars vascularis*). (c) Axial oblique FIESTA MR image reveals CN XII (black arrowhead) emerging in the preolivary sulcus, coursing lateral to the vertebral artery (black arrowhead) and directing to the hypoglossal canal. (f) Reconstructed sagittal oblique CT cisternogram image demonstrating, in descending order, the cisternal courses of CN IX, X, and XI (white arrows).



aneurysms, and trauma) may affect the ninth cranial nerve and neighboring cranial nerves (X, XI, and XII).

Vagus Nerve

Function

The main functions of the vagus nerve are phonation and swallowing. It also transmits cutaneous sensory fibers from the posterior part of the external auditory meatus, tympanic membrane, and hypopharynx. It supplies the gut tube as far as the splenic flexure of the transverse colon, and the heart, tracheobronchial tree, and abdominal viscera (Krahl & Clark, 2012).

Anatomy

The sensory fibers originate from superior and inferior ganglia situated below the jugular foramen; they reach the nuclei of the trigeminal nerve and the nucleus of the solitary tract. The efferent fibers originate from the dorsal motor nucleus of the vagus, which gives rise to preganglionic axons to innervate the heart and thoracoabdominal viscera, and from the nucleus ambiguus (branchiomotor fibers to the muscles of the pharynx and larynx) (Zurowski, Nowak, Wordliczek, Dobrogowski, & Thor, 2012). The vagus nerve arises below the glossopharyngeal nerve as a line of tightly packed rootlets posterior to the superior third of the olive; it is composed of multiple combinations of large and small rootlets that pass ventral to the choroid plexus (Rhoton, 2000b). They pass to the subarachnoid space in the cerebellomedullary cistern. The vagal rootlets enter the dural subcompartment, called the vagal meatus. They then gather in the intracranial orifice of the foramen, along the medial side of the intrajugular process, turning downward and exiting vertically, retaining an intimate relationship with the accessory nerve, behind CN IX on the posteromedial wall of the internal jugular vein and lateral to the hypoglossal nerve (Rhoton, 2000c). It descends vertically, posterior to the carotid artery and jugular vein and in the lateral pharyngeal space. It then penetrates the thoracic cavity. Here, the right nerve crosses laterally to the common carotid artery, between the subclavian artery and the jugulo-subclavian venous angle, and then laterally to the trachea; then it passes posteriorly to the right bronchi and the esophagus. The left nerve descends to the aortic arch lateral to the left common carotid artery, then crosses posterior to the left bronchi and anterior to the thoracic aorta, finally crossing anteriorly to the esophagus. Both give rise to recurrent laryngeal nerves that supply sensory and motor innervation to the larynx. The striated muscles of the soft palate, except the tensor veli palatine, pharynx, and larynx are innervated by the vagus nerve. Both nerves pass through the esophagus hiatus reaching the

stomach borders for distribution among the abdominal viscera.

The swallowing reflex is provoked when food and drink passes on to the posterior part of the tongue, and the muscles supplied by the nucleus ambiguus propel it backwards and downwards into the hypopharynx; then, through the cricopharyngeal sphincter to the esophagus. The nasopharynx is closed by the palatal muscles and the eustachian tube opens. The laryngeal orifice is reduced in size largely as a result of elevation of the entire laryngeal skeleton by the muscles attaching to it from above, and the cricopharyngeal sphincter opens. Phonation is the final step of the speaking path. The vocal cords create the narrow slit through which air is directed to make a sound; muscles that move the cords are supplied by the recurrent laryngeal nerve. The pharyngeal muscles, the tongue, the muscles of facial expression, mandibular movements, and the palate all modify the crude noise produced by the larynx to create speech. Vagal reflexes are coughing, vomiting, and fainting; irritation of the pharynx, larynx, and tracheobronchial tree can cause coughing; and irritation of nasal mucosa can cause sneezing. Vomiting is an important reflex integrated in the obex on the fourth ventricle floor. This can be conditioned by a variety of stimuli including vestibular stimulation, stimulation of the posterior pharynx wall, irritation of the esophagus and stomach mucosa, and some toxic substances.

Anatomical Landmarks

- Fourth ventricle floor
- Medullary olive
- Jugular foramen
- Internal carotid artery
- Internal jugular vein
- Trachea
- Esophagus

Figure 14.8b, e and f shows some anatomical landmarks of CN X.

Clinical Correlation

There are several lesions and reflexes associated with lesions of CN X:

1. Ipsilateral paralysis of the soft palate, pharynx, and larynx leading to dysphonia (hoarseness), dyspnea, dysarthria, and dysphagia.
2. Loss of the gag (palatal) reflex (efferent limb).
3. Anesthesia of the pharynx and larynx leading to unilateral loss of the cough reflex.
4. Aortic aneurysms and tumors of the neck and thorax that frequently compress the vagal nerve.
5. Complete laryngeal paralysis, which can be rapidly fatal if it is bilateral (asphyxia).

6. Parasympathetic (vegetative) disturbances, including bradycardia (irritative lesion), tachycardia (destructive lesion), and dilation of the stomach.
7. The oculocardiac reflex, in which pressure on the eye slows the heart rate (afferent limb of CN V₁ and efferent limb of CN X).
8. The carotid sinus reflex, in which pressure on the carotid sinus slows the heart rate (bradycardia), and the efferent limb of CN X (Fix, 2001).

Supranuclear lesions are rare; lesions within the brain stem may be due to vascular injuries, brain stem gliomas, fourth ventricle and upper cervical medulla ependymomas, syringobulbia, and motor neuron and demyelinating diseases. Tumors originated or invading the fourth ventricle floor structures develop symptoms related to compression of the vagus nerve-related nucleus early on its course. Lesions located in posterior fossa usually involve the ninth through twelve cranial nerves; lesions at this location are secondary to cranial fractures, tumors, vascular lesions, infections, and inflammatory diseases. Lesions affecting the vagus nerve on its long path can be due to neck and thorax tumors, aneurysms, or trauma; there is a risk of lesion of recurrent laryngeal nerve on neck and thorax surgery.

Surgical approaches to the ninth to the twelve nerves include posterior fossa approaches, from a retrosigmoid craniotomy, that can be extended to a far lateral approach.

Accessory Nerve

Function

The accessory nerve has two parts: cranial and spinal. The cranial part joins the vagus, from which it is functionally indistinguishable, innervating muscles of the larynx and pharynx. The spinal part is motor to the sternocleidomastoid and trapezius muscles and is not really a cranial nerve.

Anatomy

The spinal accessory part originates in the upper four or five segments of the spinal cord in the ventral horn; rootlets emerge between the ventral and dorsal spinal nerve roots, just behind the denticulate ligament, then ascend through the foramen magnum to enter the posterior cranial fossa, and briefly run with CN XI before emerging through the jugular foramen. It continues in the posterior cervical triangle crossing the internal jugular vein, posterior to the stylohyoid and digastric muscles, then reaches the deep compartment of the sternocleidomastoid and trapezius muscles (Policeni & Smoker, 2008). Supranuclear innervation of the trapezius and sternocleidomastoid muscles probably originates in the lower precentral gyrus.

The cranial root of the accessory nerve emerges below the vagus nerve in the medulla, running with the spinal part

briefly until it passes through the jugular foramen to join the vagus nerve beside the inferior vagus ganglion. Both supply the pharynx and larynx.

Anatomical Landmarks

- Medulla
- Foramen magnum
- Jugular orifice
- Posterior cervical triangle

Figure 14.8b, e and f shows some anatomical landmarks of CN XI.

Clinical Correlation

Clinical evaluation of the accessory nerve includes the motor function of the sternocleidomastoid and trapezius muscles, and sensory and motor functions of the pharynx and larynx, together with the glossopharyngeal and vagus nerves.

Lesions of CN XI are associated with:

1. Paralysis of the sternocleidomastoid muscle resulting in difficulty in turning the head to the contralateral side.
2. Paralysis of the trapezius muscle resulting in shoulder droop and inability to shrug the shoulder.
3. Paralysis of the larynx if the cranial root is involved (Fix, 2001).

Supranuclear lesions are manifested by trapezius muscle paresis and position of the head that is turned away from the hemiplegic side, because of paresis of the sternocleidomastoid muscle on the opposite side. Focal seizures in areas 8 and 9 of the cerebral cortex cause contraction of the ipsilateral sternocleidomastoid muscle, as the head turns contralateral to the epileptogenic lesion. Brain stem lesions cause similar clinical manifestations to those described earlier with vagus nerve nucleus lesions. Lesions at the subarachnoid space and jugular foramen cause lower cranial nerve syndromes, that varies depending on which cranial nerves are affected from the ninth to the twelfth. These syndromes are the Vernet, Collet-Sicard, Villaret, Schmidt, Jackson, Tapia, and Garcin syndromes. The accessory nerve can be affected from lesions in the neck, trauma, neck tumors, and iatrogenic injury being the most common causes (Iwasaki & Kondo, 1991; Laigle-Donadey, Taillibert, Martin-Duverneuil, Hildebrand, & Delattre, 2005).

Hypoglossal Nerve

Function

This nerve supplies the muscles of the tongue, which are important in chewing, in the initial stages of swallowing, and in speech.

Anatomy

The hypoglossal nerve originates from the medulla in the hypoglossal nucleus, deep to the hypoglossal triangle in the floor of the fourth ventricle; then it emerges by a vertical series of rootlets between the pyramid and olive, anterior to CNs IX, X, and XI (Monkhouse, 2006). The hypoglossal rootlets course anterolaterally through the subarachnoid space and pass behind the vertebral artery to reach the hypoglossal canal. Before entering the hypoglossal canal, the rootlets collect into two bundles; after passing through the canal, these bundles unite and the nerve lies medial to the internal jugular vein and the glossopharyngeal, vagus, and accessory nerves (Rhoton, 2000b). In the cranial base it descends by the lateropharyngeal space, anteriorly and inferiorly, posteromedial to the internal carotid artery. Then it passes laterally to the external carotid artery, just inferior to the origin of the occipital artery, and then reaches the suprahyoid cervical region, where it is situated laterally to the hyoglossus muscle, which separates it from the lingual artery. Finally, it joins the submandibular canal to reach the tongue muscles. Supranuclear control of the tongue is mediated by corticobulbar fibers that originate mainly within the lower portion of the precentral gyrus area.

Anatomical Landmarks

- Fourth ventricle floor
- Medulla pyramid
- Occipital condyles
- External carotid artery
- Suprahyoid cervical region

Figure 14.8c shows the cisternal course of CN XII.

Clinical Correlation

Lesions of CN XII result in specific clinical findings: transection of CN XII results in hemiparesis of the tongue; protrusion causes the tongue to point toward the weak side because of the unopposed action of the opposite genioglossus muscle (Fix, 2001).

Supranuclear lesions that affect the corticobulbar tract anywhere in its course may result in tongue paralysis; unilateral lesions of hypoglossal nucleus or nerve result in paresis, atrophy, and fasciculation that affect half of the tongue, and because of the proximity of both hypoglossal nuclei, lesions in this area often results in bilateral affection.

Because of its close proximity in the posterior fossa, CN XII often results in effects within lesions that affect the ninth to eleventh cranial nerves, but it may be affected solely in condylar fractures. It can also be solely affected in the neck or in its more distal course near the tongue. Causes of this peripheral involvement include carotid aneurysms, vascular entrapment, dissection of the internal

carotid artery, infections, trauma, neck tumors, and iatrogenic injury.

CONCLUSIONS

Imaging of cranial nerves requires a thorough understanding of neuroanatomy and neurophysiology. Current challenges in CN imaging include finding an adequate tailoring of imaging studies by balancing the techniques needed to examine a presumptive anatomical area of interest within a reasonable time frame. A topographical diagnosis supported by neurological and electrophysiological testing can be very helpful in allowing imaging studies to focus on specific segments of the CN. As CT and MR imaging grow more sophisticated and more dedicated, there will be an increasing demand for the specialist's knowledge of these subjects to support clinical diagnosis.

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